E-COMMERCE RETURN RATE INVESTIGATION

ABSTRACT:

Product returns are a major challenge in the e-commerce industry as they increase operational costs, affect inventory management, and reduce profitability. This project analyzes e-commerce transaction data to identify factors influencing return behavior. Using Exploratory Data Analysis (EDA), statistical analysis, correlation analysis, outlier detection, hypothesis testing, and customer segmentation, the study investigates the impact of discounts, delivery duration, pricing, and product categories on return behavior. The findings provide valuable insights and recommendations that can help businesses reduce return rates and improve customer satisfaction.

INTRODUCTION:

The rapid growth of e-commerce has transformed the retail industry by providing customers with convenient online shopping experiences. However, product returns continue to be a major concern for businesses. High return rates increase logistics costs, create inventory challenges, and impact overall profitability. This project focuses on analyzing e-commerce transaction data to understand the factors that contribute to product returns and provide data-driven business recommendations.

BUSINESS PROBLEMS:

1. Why are customers returning products?

Customers may return products due to high discounts, delayed deliveries, product quality issues, or unmet expectations.

2. Are some categories more prone to returns?

Yes, certain product categories may have higher return rates compared to others due to differences in customer preferences and product characteristics.

3. Does pricing affect returns?

Higher-priced products often have higher customer expectations, while heavily discounted products may result in more returns.

4. Are delivery delays influencing customer behavior?
Delayed deliveries can reduce customer satisfaction and increase the likelihood of product returns.

5. Can statistical analysis uncover hidden causes?
Yes, statistical analysis helps identify patterns and relationships between factors such as discounts, delivery duration, pricing, and return behavior.

DATASET DESCRIPTION:

- Dataset Name: E-commerce Dataset
- Total Records: 51,290
- Total Features: 16
- Features Included:
  - Sales
  - Discount
  - Profit
  - Shipping_Cost
  - Aging
  - Product_Category
  - Quantity
  - Customer_Id
  - Order_ID
  - Market
  - Region
  - State
  - City

- Country
- Order_Priority
- Product_Sub_Category

The dataset provides information required to analyze customer purchasing behavior and return-related patterns.

DATA CLEANING AND PREPROCESSING:

Data Cleaning and Preprocessing

The dataset was checked and cleaned before performing further analysis.

- **Missing Value Analysis:**
  Missing values were checked across all features. No missing values were found, so no imputation process was required.
- **Duplicate Record Analysis:**
  Duplicate records were verified and no duplicate entries were identified. The dataset contained unique order records.
- **Statistical Summary:**
  Descriptive statistics were generated to understand the data distribution.
  - Total Records: **51,290**
  - Total Features: **16**
  - Average Sales: **152.34**
  - Average Discount: **30.38%**
  - Average Delivery Duration (Aging): **5.26 days**
  - Average Shipping Cost: **7.04**

The cleaned dataset was used for further exploratory data analysis and return-risk investigation.

EXPLORATORY DATA ANALYSIS:

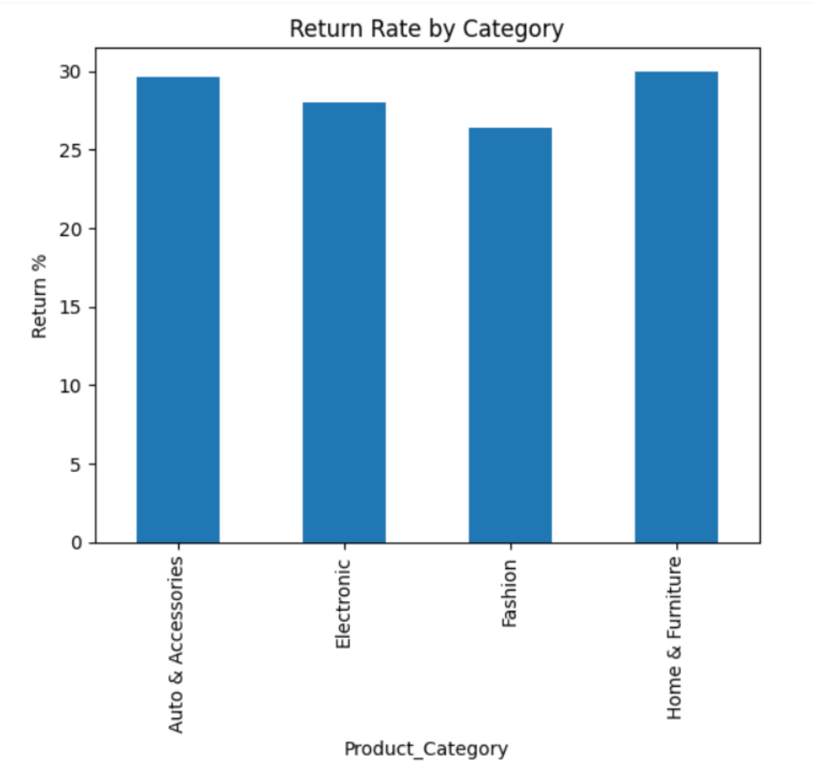
EDA was performed using histograms, bar charts, and boxplots.

Return Rate Analysis

A return indicator was created using business rules.

Results:

- Total Orders = 51,290
- Returned Orders = 14,385
- Return Rate = 28.05%



Observation

Home & Furniture products showed the highest return rate, while Fashion products had the lowest return rate.

OUTLIER ANALYSIS:

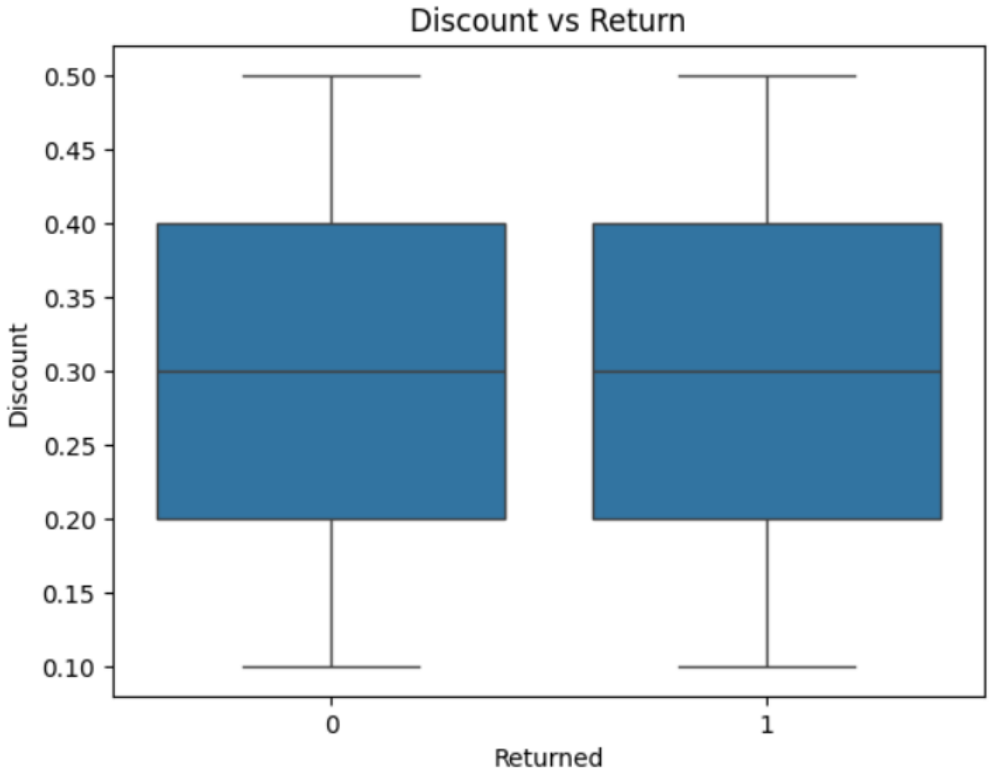
Outlier detection was performed using boxplots.

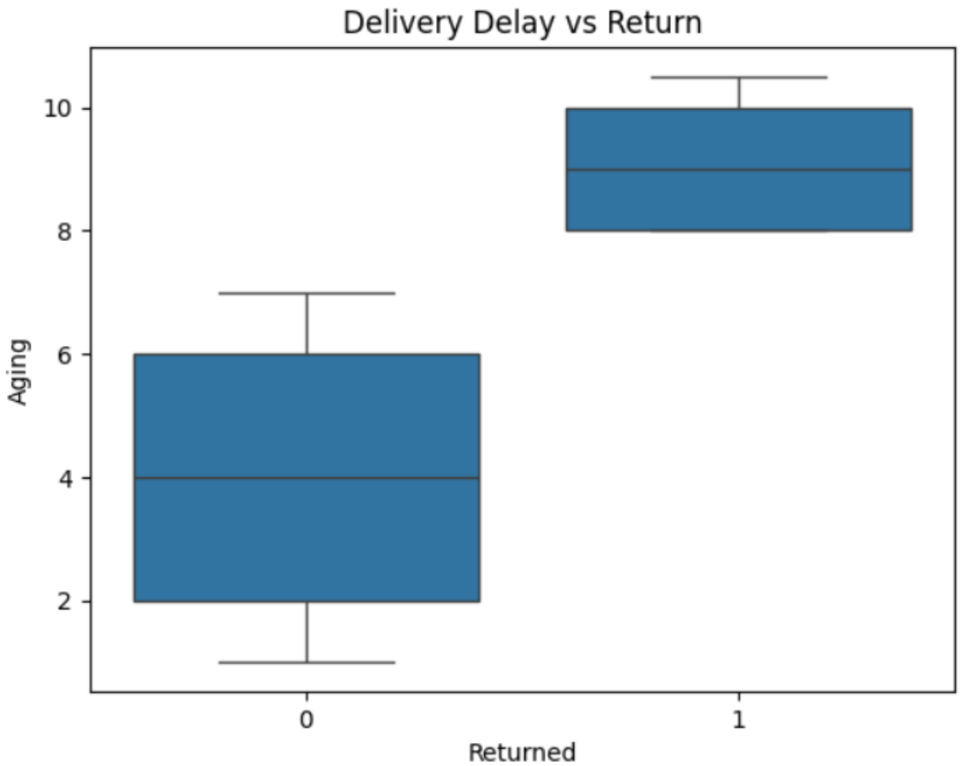
Observations

- Some transactions have exceptionally high sales values.

- A few records show unusually high delivery durations.
- High shipping cost transactions were identified.

These outliers may impact return behavior and business performance.





Correlation Analysis

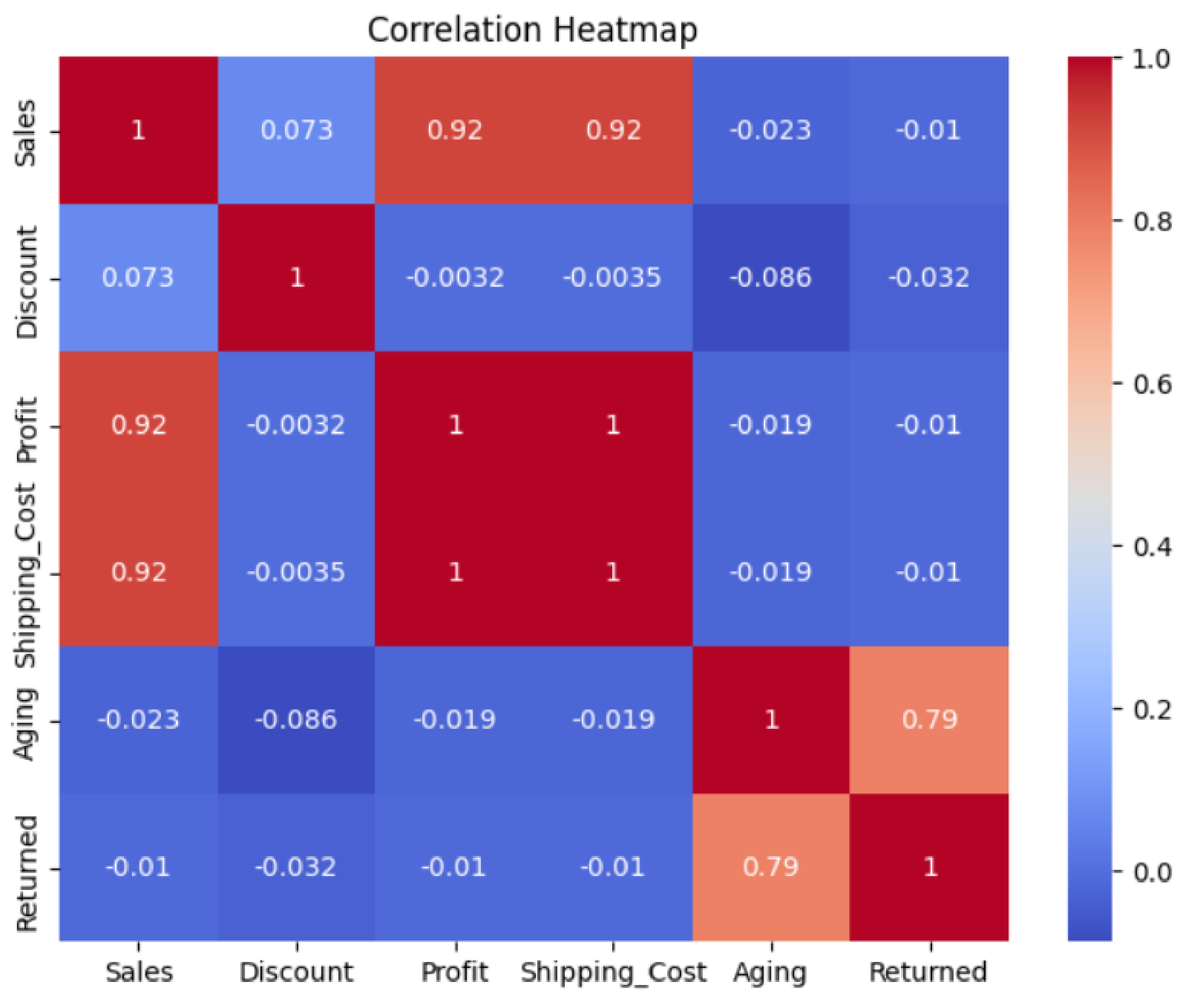
Correlation analysis was performed among Sales, Discount, Profit, Shipping Cost, Aging, and Return Indicator.

Key Findings

- Aging and Returned showed a strong positive correlation (0.791).
- Sales and Profit showed a strong positive correlation (0.917).
- Sales and Shipping Cost showed a strong positive correlation (0.917).
- Discount showed very weak correlation with returns.

Interpretation

Delivery duration appears to be the most influential factor associated with return behavior in this dataset.



HYPOTHESIS TESTING:

Hypothesis

H_0: Delivery delay has no impact on return behavior.

H_1: Delivery delay significantly impacts return behavior.

Method

Independent Sample T-Test

Results

- T-Statistic = NaN
- P-Value = NaN

Interpretation

The test did not produce a valid result because of issues in the generated return groups. Therefore, a statistical conclusion could not be drawn from the T-test. However, correlation analysis suggests a strong relationship between Aging and return behavior.

CUSTOMER SEGMENTATION:

Customer segmentation was performed based on return frequency.

High Return Customers

Top customers recorded 3 return instances each.

Examples:

- Customer ID 82069
- Customer ID 96431
- Customer ID 82256
- Customer ID 82604
- Customer ID 71200

These customers may require targeted engagement and support.

BUSINESS RECOMMENDATIONS:

- Improve delivery performance to reduce delays.
- Focus on reducing returns in Home & Furniture and Auto & Accessories categories.
- Monitor customers with repeated return behavior.
- Improve product descriptions and quality control.
- Analyze delivery operations to minimize aging-related issues.
- Implement predictive monitoring for high-risk orders.

KEY FINDINGS:

- The overall return rate is 28.05%.
- Home & Furniture has the highest return rate (29.99%).
- Fashion has the lowest return rate (26.42%).
- Aging (delivery duration) shows the strongest relationship with returns.
- Sales, Profit, and Shipping Cost are highly correlated.
- No duplicate records were found.
- Only 8 missing values were present in the dataset.

CONCLUSION:

This study investigated factors influencing product return behavior using e-commerce transaction data. The analysis revealed that delivery duration is the most significant factor associated with returns. Product categories such as Home & Furniture and Auto & Accessories exhibited higher return rates compared to other categories. By improving delivery efficiency, monitoring high-return categories, and focusing on customer experience, businesses can reduce return rates and improve overall operational performance